

The Economics of Mine Waste Reduction

The Case of Ore-Sand

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Outline

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3. Calibration & simulation
4. Private economic benefits of adoption
5. Adoption amid cost and revenue shocks
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Introduction

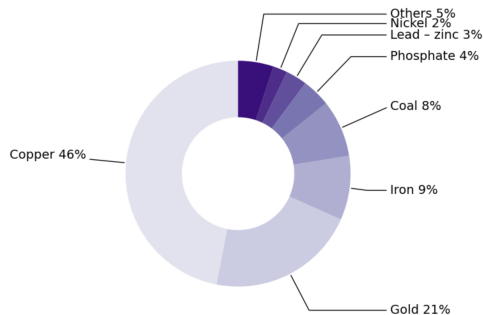
Three converging pressures

- Rising mineral demand meets declining ore grades → waste volumes rise sharply.
- Tailings are often a mine's most significant environmental liability.
- Tailings-management market $\approx$ US\$15bn (2025), growing $\approx$ 7%/yr.

Why it matters

- Global tailings production: $\sim$ 14 billion tonnes/yr; cumulative storage $>$ 200 billion tonnes.
- Major dam failures (Brumadinho 2019, Samarco 2015) caused US\$5–55B in damages each.
- Regulatory pressure increasing: Global Industry Standard on Tailings Management (GISTM, 2020) raises compliance costs.

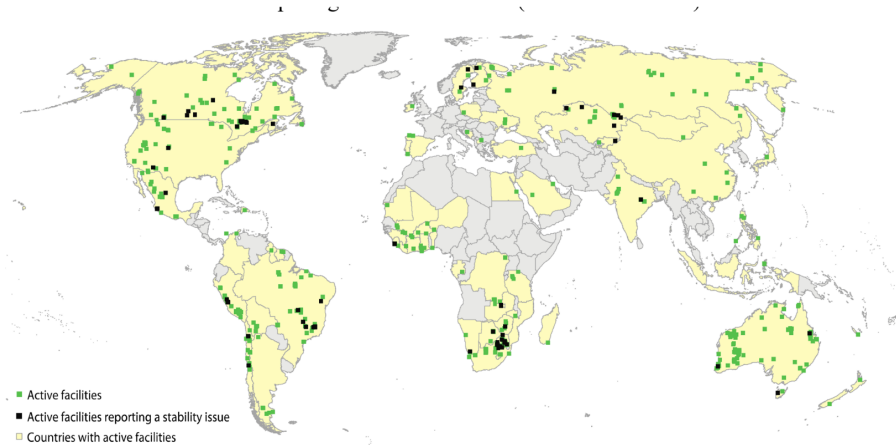
The scale of mine waste



Copper accounts for **46%** of all recorded tailings dam failures.

- The green energy transition is accelerating mineral demand — copper, lithium, nickel, cobalt.
- Declining ore grades mean more material must be mined per unit of metal recovered.
- Result: **30–60 billion tonnes** of mine waste produced annually, and growing (Islam & Murakami, 2021).
- Waste rock (72,000 Mt) + tailings (8,850 Mt) $\gg$ ore extracted (10,180 Mt) (Baker et al., 2020).
- Tailings-management market $\approx$ US\$15bn (2025), growing $\approx$ 7%/yr.

The global footprint of tailings storage



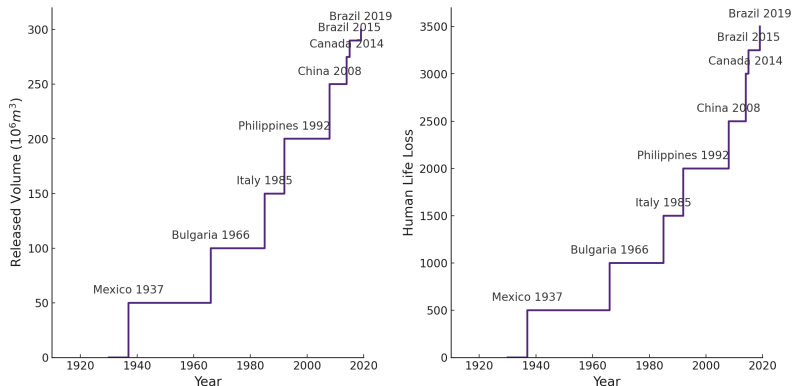
Active tailings facilities worldwide (Franks et al., 2021). Black squares indicate facilities reporting stability issues. Regulatory tightening via the Global Industry Standard on Tailings Management (GISTM, 2020).

Tailings facilities



Tailings storage facilities at mines worldwide (Werner et al., 2020). The environmental footprint of tailings often exceeds that of the mine pit itself.

Tailings failures



Cumulative released volume and human life loss from major tailings dam failures, 1915–2020 (Islam & Murakami, 2021). **Brumadinho (2019)**: 270 deaths, US\$5–55B in claims. **Samarco (2015)**: 19 deaths, 60M m^3 released. Climate change exacerbates failure risks (Santamarina et al., 2019); financial bonds and rehab guarantees increasingly required.

Tailings failures



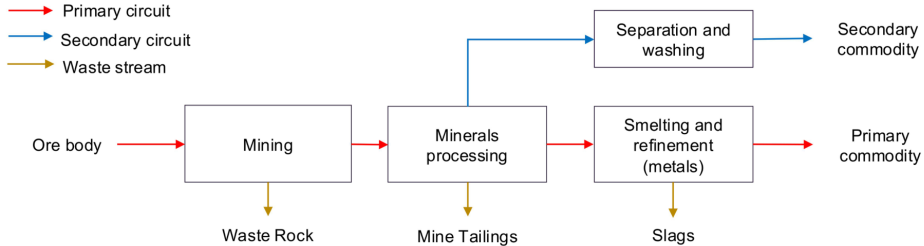
Brumadinho, Brazil (2019). Córrego do Feijão tailings dam collapse sequence. Robertson et al. (2019).

- 270 fatalities
- 12 million m³ released
- Claims: US\$5–55 billion
- Comparable to the entire operation's NPV

Any technology that reduces tailings volume reduces both recurrent costs and catastrophic risk.

(Santamarina et al., 2019; Robertson et al., 2019; Franks et al., 2021)

Ore-sand



Ore-sand is recovered from within existing mineral processing circuits, materials otherwise sent to tailings storage are diverted into a manufactured sand product with construction applications (road sub-base, concrete aggregate, bricks).

This paper

- Develops and calibrates an economic model to quantify the private benefits of ore-sand adoption.
- Captures the complexity of medium-to-large mining operations: stochastic prices, declining ore grades, production complementarities, waste costs.
- Uses Monte Carlo simulation to evaluate profitability across adoption scenarios.

Three contributions:

1. First quantification of ore-sand profitability as a cleaner production technology.
2. Economic model integrating waste costs (fixed & variable) into the mine-investment decision.
3. Framework designed to be commodity-general, calibrated to iron ore.

Model overview

1. **Two products:** primary commodity (iron ore) + secondary (ore-sand from tailings).
2. **Production:** Cobb–Douglas technology ($K^\alpha L^\beta M^\gamma$) for each product, aggregated via CES (low substitutability, secondary is complement to primary).
3. **Costs:** iceberg-type revenue retention (τ) captures processing, waste management, and transport. Waste costs have variable and fixed (rehabilitation) components.
4. **Adoption benefit:** ore-sand reduces waste costs through both channels (ω = waste reduction factor), raising τ .
5. **Prices:** geometric mean-reverting stochastic process for both commodities.
6. **Dynamics:** capital accumulation, ore-reserve depletion, declining ore grades over mine life.
7. **Solution:** Monte Carlo simulations; NPV comparison of adoption vs BAU across five scenarios (varying commercialisation and rehabilitation assumptions).

Annual profits

The mine operator aggregates primary (p) and secondary (s) production:

$$\pi_t = \underbrace{\tau_p p_t^p q_t^p + \tau_s p_t^s q_t^s}_{\text{net revenue}} - \underbrace{(VC_t^p + VC_t^s)}_{\text{variable costs}} - \underbrace{(\varphi^p + \varphi^s)}_{\text{fixed costs}} - \underbrace{\varphi^r}_{\text{rehab costs}}$$

- τ_j = revenue-retention parameter (“iceberg” cost share retained after transport, processing, waste)
- p_t^j = output price of commodity j at time t
- q_t^j = quantity produced of commodity j
- VC_t^j = variable factor costs (capital, labour, materials)
- φ^j = fixed costs; φ^r = mine closure and rehabilitation costs

Production technology

Aggregate output combines primary and secondary via a CES function:

$$Q = \mu [\delta_p q_p^\rho + \delta_s q_s^\rho]^{1/\rho}, \quad \varepsilon = \frac{1}{1 - \rho}$$

- Low substitutability: primary and secondary are complements in production, not substitutes. Secondary output is constrained by (and proportional to) primary.

Each process $j \in \{p, s\}$ uses a Cobb–Douglas technology:

$$q_j = a_j \varepsilon_j \nu_j K_j^\alpha L_j^\beta M_j^\gamma, \quad \alpha + \beta + \gamma = 1$$

- a_j = recovery rate; ε_j = processing efficiency; ν_j = ore grade index

Material inputs

Material inputs M are decomposed into energy (E) and ore inputs (R) via a nested Cobb–Douglas:

$$M_j = E_j^\eta R_j^{1-\eta}$$

Production path over mine life:

$$Q_t = Q_0 (1 + g)^t e^{-\delta_j(t-t_{\text{peak}})^2}$$

- g = production growth rate
- δ_j = decline parameter (secondary declines faster at mine end)
- t_{peak} = peak production year

Cost structure

Total costs for each commodity j :

$$C_j = \underbrace{r K_j + w L_j + p^M M_j}_{\text{variable factor costs}} + \underbrace{\varphi_j}_{\text{fixed costs}}$$

where $p^M = p_\eta^E p_{1-\eta}^R$ is a composite material price (energy price p^E , ore price p^R).

Iceberg costs:

$$\tau_j = 1 - c_{\text{proc},j} - c_{\text{waste},j} - c_{\text{trans},j}$$

Waste costs have two components:

1. Variable: $c_{\text{waste},j}$: proportional to production (iceberg-type discount on revenue)
2. Fixed: φ^r : mine closure and rehabilitation costs (known at operation start, based on total ore resources)

Waste cost reduction under adoption

Ore-sand reduces waste costs through both channels:

Variable waste cost of secondary production acts as a *credit*:

$$c_{\text{waste}}^{(s)} = -\omega c_{\text{waste}}^{(p)}, \quad \omega \in [0, 1]$$

Rehabilitation costs reduced proportionally:

$$\varphi_{\text{adopt}}^r = (1 - \omega) \varphi_{\text{BAU}}^r$$

- Revenue retained rises: $\tau_{\text{adopt}} > \tau_{\text{BAU}}$

τ is the key parameter: it captures the share of revenue retained after all iceberg costs.
Lower $\tau \Rightarrow$ thinner margins $\Rightarrow$ stronger leverage from waste-cost savings.

Extensive and intensive margins of adoption

Extensive margin (adopt or not): adoption requires that total variable revenue over the mine life covers the fixed costs of secondary production:

$$\sum_{t=0}^T [\tau_s p_t^s q_t^s - VC_t^s] e^{-\delta t} \geq \varphi^s$$

Intensive margin (how much to produce): determined by relative marginal costs. The zero-profit condition provides the threshold:

$$\frac{\partial \pi}{\partial q_s} = \tau_s p_t^s - MC_t^s = 0 \quad \Rightarrow \quad q_s^* \text{ where } p^s = \frac{MC^s}{\tau_s}$$

- Entry is driven by fixed cost recovery (extensive) — waste-cost savings help here
- Scale is driven by marginal cost comparison (intensive)
- Distinguishes *co-product* (secondary essential for profitability) vs *by-product* (secondary not essential)

Dynamics of the mining operation

Value function (dynamic programming):

$$V(s_t) = \max_{K_t, L_t, M_t} \{ \pi(s_t) + \beta \mathbb{E}[V(s_{t+1}) | s_t] \}$$

where s_t = state (prices, capital, reserves, ore grade) and $\beta = e^{-\delta}$ is the discount factor.

Capital accumulation:

$$K_{t+1} = (1 - d) K_t + I_t - \text{adj}(I_t)$$

Ore reserve depletion:

$$R_{t+1} = R_t - Q_t$$

Ore grades evolve exogenously, indexed by mine age (declining over time).

Output price volatility

Commodity prices follow a geometric mean-reverting stochastic process:

$$dP_t = \kappa (\ln \bar{P} - \ln P_t) P_t dt + \sigma P_t dW_t$$

- $\bar{P}$ = long-run average price; κ = mean-reversion speed
- σ = price volatility; dW_t = Brownian motion
- Prices naturally revert to $\bar{P}$ at speed κ

Expected future price:

$$\mathbb{E}[P_{t+h} | P_t] \rightarrow \bar{P} \quad \text{as } h \rightarrow \infty \quad (\text{adjustment speed } \kappa)$$

Consistent with observed mineral commodity price behaviour (Schwartz 1997; Cortazar & Casassus 1998; Bernard et al. 2008).

After-tax cash flows & NPV

After-tax cash flow in period t under adoption:

$$CF_t = (\tau_p p_t^p q_t^p + \tau_s p_t^s q_t^s - VC_t^p - VC_t^s - \varphi^p - \varphi^s)(1 - \tau_{\text{tax}}) - (1 - \omega) \varphi^r$$

Expected NPV over $N = 1,000$ Monte Carlo simulations:

$$\mathbb{E}[\text{NPV}] = \frac{1}{N} \sum_{i=1}^N \sum_{t=0}^T CF_{i,t} e^{-\delta t}$$

- Each simulation draws a stochastic price path for both commodities
- Optimal factor demands computed via backward induction
- Two scenarios per path: adoption vs BAU (primary only)

Quantification

Step 1 — Calibration:

- Representative iron ore mine (Vale/Brucutu, Brazil): 30-year mine life, 3 Mt/yr, +3% growth.
- Parameters from S&P Global Mine Economics, company reports, and engineering literature.
- Production path, cost structure (τ), and fixed costs matched to observed data.

Step 2 — Simulation:

- Monte Carlo draws of stochastic price paths (mean-reverting, $\sigma = 5\%$, $\kappa = 0.65$).
- For each draw: solve optimal factor demands via backward induction; compute after-tax cash flows under BAU and each adoption scenario.
- NPV at 5% discount rate (robustness checks at 3%, 7%, 10%).

Step 3 — Sensitivity:

- Sensitivity over key parameter pairs: waste/rehab costs, transport/processing, technology (λ , ε), output prices, price volatility.

Calibration

The model is calibrated to a representative large-scale iron ore operation (Vale/Brucutu, Brazil) using S&P Global Mine Economics data. Low substitutability between primary and secondary output ($\varepsilon = 0.33$) reflects that ore-sand is a complement to, not a substitute for, primary production.

Table 1: General parameters

Parameter	Value	Source
Elasticity of substitution (ε)	0.33	Agera (ore-sand company)
Depreciation rate (d)	0.10	Gross & Hansen (2013)
Tax rate (τ_{tax})	0.35	S&P
Discount rate (δ)	0.05	Armstrong et al. (2019)
Growth rate (g)	0.03	S&P
Mine life (T)	30 yr	Armstrong et al. (2019)

NPV evaluated at 5% baseline discount rate, with robustness checks at 3%, 7%, and 10%. The 35% corporate tax rate reflects the Brazilian fiscal regime for mining.

Production, cost & price parameters

Table 2: Production structure, iceberg costs, and price process

Parameter	Primary (iron ore)	Secondary (ore-sand)	Source
<i>Production structure</i>			
Relative intensity (δ_j)	0.95	0.05	Agera
$\alpha / \beta / \gamma$	0.30 / 0.40 / 0.30	0.30 / 0.40 / 0.30	S&P
Energy intensity (η)	0.80	0.80	S&P
<i>Iceberg costs (share of revenue)</i>			
Transport (c_{trans})	0.30	0.015	S&P
Processing (c_{proc})	0.15	0.008	S&P
Waste mgmt (c_{waste})	0.10	−0.05	Agera
<i>Price process</i>			
Initial price (USD/dmt)	100	8	Golev et al. (2022)
Long-run mean ($\bar{P}$, USD/dmt)	120	10	Golev et al. (2022)
Volatility (σ)	0.05	0.05	Armstrong et al. (2019)
Mean reversion (κ)	0.65	0.65	Armstrong et al. (2019)

Initial values & fixed costs

Table 3: Initial production, inputs, prices, and fixed costs

Variable	Primary (iron ore)	Secondary (ore-sand)	Source
<i>Initial production</i>			
Production (dmt)	3.0×10^6	1.2×10^6 (40%)	S&P
<i>Initial inputs</i>			
Capital (USD)	479×10^6	24×10^6 (5%)	S&P
Labour (hrs/yr)	6.7×10^6	0.33×10^6 (5%)	S&P
Ore inputs (mt)	60.0×10^6	3.0×10^6 (5%)	S&P
Energy (kWh)	153.1×10^6	7.7×10^6 (5%)	S&P
<i>Input prices</i>			
Cost of capital (USD/unit)	0.1	0.1	Cox et al. (2022)
Wage (USD/hr)	12.72	12.72	S&P / Brucutu
Ore inputs (USD/t)	0.62	0.62	S&P
Energy (USD/kWh)	0.14	0.14	S&P / Brucutu
<i>Fixed costs</i>			
Fixed costs (USD)	2.0×10^9	0.03×10^9	S&P / Agera
Rehabilitation (USD)	2.0×10^9	—	Assumed

Adoption scenarios

Scenarios vary along two dimensions: whether ore-sand is commercialised and whether rehabilitation costs are avoided (reduced proportionally to waste diversion).

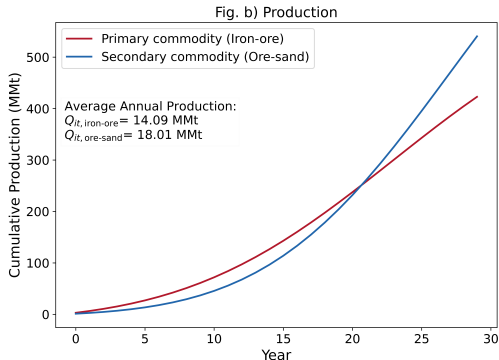
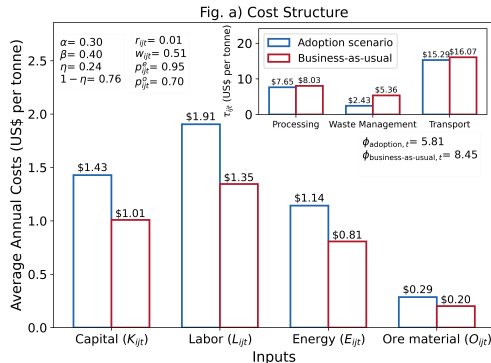
Table 4: Adoption scenarios

Scenario	Label	Definition
BAU	—	No ore-sand (primary only)
Base Adoption	Base	50% waste reduction ($\omega = 0.5$), commercialised
S2	Not comm./full rehab	Ore-sand given away, full rehabilitation costs
S3	Not comm./no rehab	Ore-sand given away, rehabilitation costs avoided
S4	Comm./full rehab	Ore-sand sold, full rehabilitation costs
S5	Comm./no rehab	Ore-sand sold, rehabilitation costs avoided

S5 is the most favourable scenario, it captures both ore-sand revenue and avoided rehabilitation. S2 is the most conservative, it isolates the pure waste-cost savings channel. Even S2 can be profitable if waste-cost savings are large enough relative to the incremental investment.

Cost structure & production

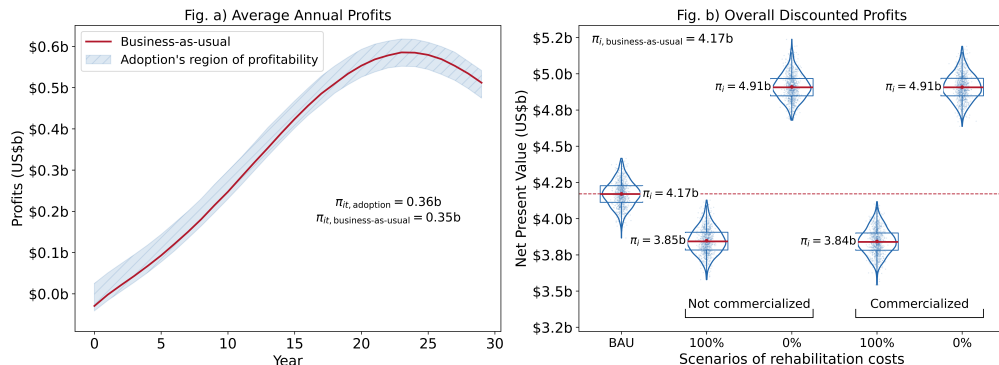
Figure 1: BAU vs adoption over 30-year mine life (1,000 MC simulations)



(a) Adoption lowers average unit costs (US\$5.81/t vs US\$8.45/t per tonne processed). Waste-management costs fall from US\$5.36/t to US\$2.43/t. (b) Cumulative ore-sand exceeds iron ore output (~ 18 vs ~ 14 MMt/yr).

Economic benefits of adoption

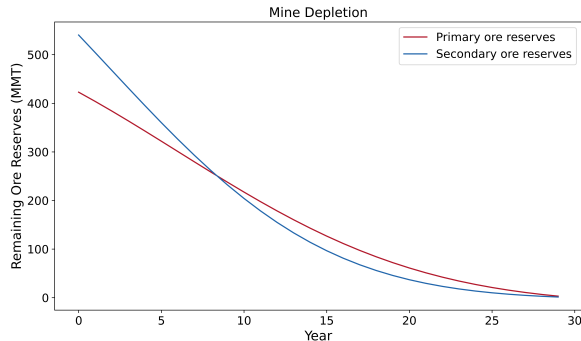
Figure 2: Average annual profits and NPV distribution across scenarios



(a) Adoption profit: US\$0.36B/yr vs BAU US\$0.35B/yr, with 95% CI. (b) NPV 5%: BAU US\$4.17B; S5 (comm./no rehab) US\$4.91B (+18%). Even S2 (not comm./full rehab) yields US\$3.85B.

Ore reserve depletion

Figure 3: Ore reserve depletion

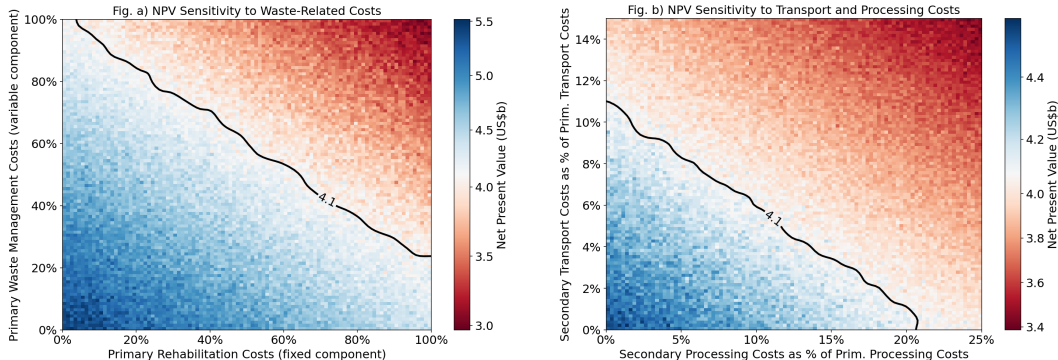


- Primary reserves: ~425 MMT initial
- Secondary reserves: ~540 MMT initial
- Both deplete to zero by year 30
- Secondary depletes faster (higher extraction rate after year ~9)
- Ore-sand is measured in equivalent units of iron ore

Secondary ore reserves are derived from primary tailings. Ore-sand production does not deplete primary reserves.

NPV sensitivity to waste & transport costs

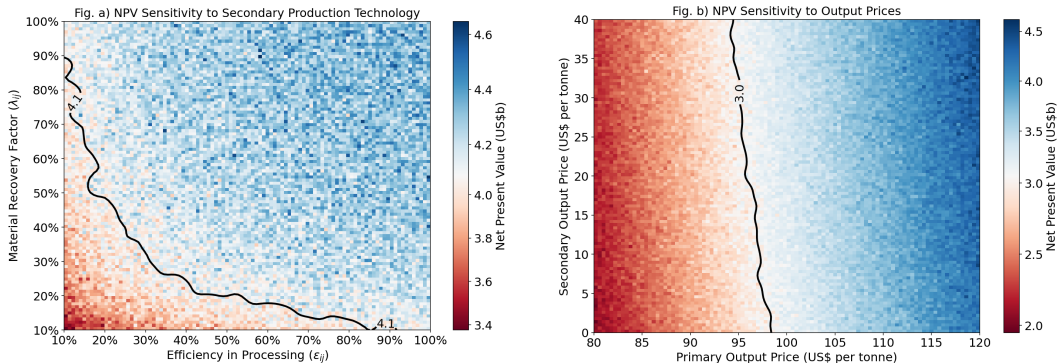
Figure 4: NPV across waste-related and transport/processing cost parameters



(a) ~50% reduction in both fixed (rehab) and variable (waste-mgmt) costs yields NPV > BAU. The iso-NPV contour shows the adoption threshold. (b) Secondary transport and processing costs have modest effects relative to waste savings.

NPV sensitivity to technology & prices

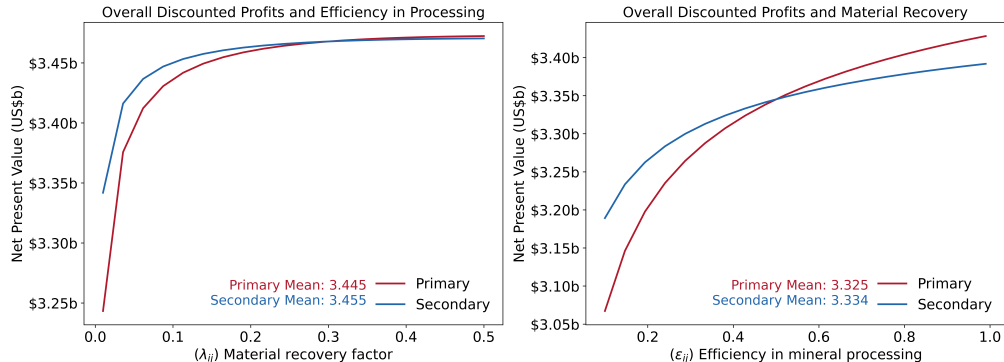
Figure 5: NPV across secondary production technology and output prices



- (a) Recovery (λ_{ij}) and efficiency (ε_{ij}) are complementary, both must exceed moderate levels.
- (b) NPV driven overwhelmingly by primary commodity price. Ore-sand price (\$10/t) has negligible impact.

Processing efficiency and material recovery

Figure 6: NPV and processing efficiency / material recovery



Diminishing returns at high values. Both parameters exhibit concavity, initial improvements yield the largest NPV gains.

Summary of main findings

1. Ore-sand adoption generates substantial additional profits vs BAU (5% to 18%), mediated by reductions in tailings management, storage, and rehabilitation costs.
2. Economic incentives shape both the extensive margin (adoption decision) and the intensive margin (scale of co-production), and remain robust to uncertainty in prices, ore grades, recovery rates, and processing costs.
3. The primary economic incentive is cost savings and constraint relief, not ore-sand revenue. Ore-sand price has negligible impact on NPV.
4. A minimum 40% reduction in waste costs is sufficient to match BAU profitability. Beyond this threshold, every additional percentage point adds to NPV.

Limitations

The model is a conservative lower bound on the value of adoption. Three channels are deliberately omitted (all favour adoption):

1. Tailings-failure risk: claims from major failures (US\$5–55B) comparable to operation NPV. Even small risk reduction yields large expected savings (Armstrong et al. 2019: ~20% NPV gain when failure risk included).
2. Mine-life extension: where tailings storage is the binding constraint, waste reduction could unlock additional primary production years.
3. Social & environmental externalities: reduced long-term environmental liabilities (acid mine drainage, seepage, dust), displaced environmentally damaging aggregate extraction (river/coastal sand mining).

The model also uses a simplified waste-cost structure (fixed + variable) and does not capture discrete tailings storage facility investments over mine life.

Conclusions

- Ore-sand co-production is a privately profitable cleaner production technology for mining.
- Value comes from waste-cost savings and higher primary output, not ore-sand revenue.
- Results robust across discount rates, adoption scenarios, and commodity price paths.
- The framework is designed to be commodity-general: it requires context-specific calibration, not fundamental model restructuring.

Future directions:

- Tailings-capacity constraints as production constraints; mine-life extension modelling.
- Social cost-benefit analysis with monetised externalities.
- Sector-wide adoption and general equilibrium effects on aggregate markets.
- Policy instruments: green mining bonds, tax incentives, streamlined permitting.